

# RAHUL MITRA

Singapore · mitrarahul2002@gmail.com · linkedin.com/in/rahulmitra-dev · github.com/Rah-Rah-Mitra · rahul-mitra.com

## EDUCATION

### National University of Singapore

Singapore

*Bachelor of Engineering, Industrial Systems Engineering*

Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.

## EXPERIENCE

### STMicroelectronics

Singapore

*Operations Research (NUS System Design Project), Contract*

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

### Amazon

Singapore

*Robotics Vision Engineer (Blended AI+X)*

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.

### Abbott Laboratories

Singapore

*Operational AI Systems & Data Engineer (Contract)*

Jun 2026 – Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

### Abbott Laboratories

Singapore

*Data Analytics and Project/Product Management Intern*

Jan 2026 – Jun 2026

- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of unseen raw data with zero execution failures.
- Productionized a team-built APC simulator for live internal users on Azure App Service, applying access restrictions, runtime and session hardening, and Docker packaging.

### YesWeHack

Remote

*Independent Bug Bounty Researcher*

May 2024 – Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports using Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.

### Republic of Singapore Navy

Singapore

*Base Support Assistant*

Jan 2022 – Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.

## PROJECTS

### AsyncDDGS — Open-Source Async Search Client (PyPI)

2024 – Present

- Authored and maintain an asyncio-first, aiohttp-based DuckDuckGo client on PyPI with pytest CI/CD, adopted across AI tooling, Discord bots, and search projects.

### Arcane Security Tooling Family

Mar 2026

- Built Rust-forward security and red-team tooling (Arcane, Arcane-PP, Arcane-GLM, Arcane-OCR) for automation, OCR, and adversarial workflows.

### DSTA BRINHACK 2025 CTF (top quartile of 400+ teams)

2025

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure in a first CTF, finishing in the top quartile among 400+ teams.

## LEADERSHIP AND ACTIVITIES

### NTUC Health Active Ageing Centre

Singapore

*Elderly Care Volunteer*

Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

## SKILLS AND CERTIFICATIONS

**Technical:** Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, fuzzing, reverse engineering, custom exploit scripting, Python, Bash, Rust

**Certifications:** Reverse Engineering Windows Executables (digital forensics); Docker for the Absolute Beginner (Stone River eLearning)